Assignment I

DNNs

January 26, 2026

Abstract

The goal is to review the basics of automatic differentiation and Pytorch. Please upload your solutions to Classroom before TBA.

Problem 1

Let $\mathbf{x}$ and $\mathbf{1} := (1, 1, \cdots, 1)$ lie both in $\mathbb{R}^n$. Consider the Rosenbrock's function:

$$f(\mathbf{x}) = \sum_{i=1}^{n-1} \Big[100(x_{i+1} - x_i^2)^2 + (x_i - 1)^2 \Big]. \tag{1}$$

Compute $\nabla f(\mathbf{1})$ using reverse mode automatic differentiation (by hand) for $n = 5$.

Problem 2

Make a Python script to compute the eigenvalue with maximum modulus and its corresponding eigenvector of a symmetric matrix $A \in \mathbb{R}^{n \times n}$. Use Pytorch instead of Numpy/Scipy packages. Use the Power method, described (for instance) in [1], in section 8.2.1. Next, test your program with the following matrix

$$\begin{pmatrix} -1.6407 & 1.0814 & 1.2014 & 1.1539 \\ 1.0814 & 4.1573 & 7.4035 & -1.0463 \\ 1.2014 & 7.4035 & 2.7890 & -1.5737 \\ 1.1539 & -1.0463 & -1.5737 & 8.6944 \end{pmatrix}.$$

Problem 3

Make a Python script using Pytorch to minimize function (1) for $n = 10$ using Newton's method. Plot how the error decreases as the algorithm iterates.

Problem 4

Make a Python script using Pytorch to minimize function (1) for $n = 10$ using BFGS's method. Plot how the error decreases as the algorithm iterates.

Problem 5

Make a Python script using Pytorch to minimize function (1) for $n = 10$ using Adam's method. Plot how the error decreases as the algorithm iterates.

References

[1] G.H. Golub, C.F. Van Loan, **Matrix Computations**, The Johns Hopkins University Press, 1996.